*Sec :* *Date :*

# SRI CHAITANYA EDUCATIONAL INSTITUTIONS,INDIA.

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**SEC :Out going srs and longterm** **DATE: 24.07.2020**

**SUB:** **NEET GRAND TEST-14** **Max. Marks : 720**

***IMPORTANT INSTRUCTIONS :***

1. **The question paper consists of 4 parts (Part-I Physics, Part-II Chemistry, Part-III Botany and Part- IV Zoology). Each part contains 45 Questions.**
2. **Each question has 4 choices (a), (b), (c) and (d), out of which only ONE is correct.**
3. **For each correct answer + 4 MARKS, Wrong Answer -1 Mark.**
4. **Use BLUE BALL POINT PEN only to darken the appropriate circle.**

## PHYSICS

01. A simple pendulum is vibrating with an angular amplitude of 90°. For what value of $\theta$ is the acceleration directed

i. vertically upward

ii. horizontally

iii. vertically downward

a. $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right), 0°, 90°$

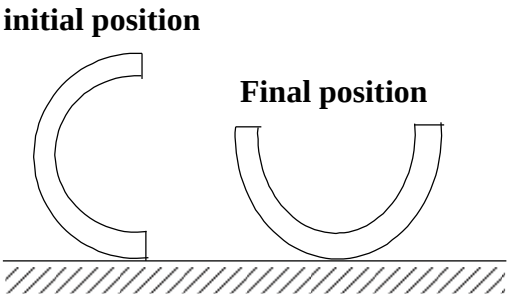
b. $0°, \cos^{-1}\left(\frac{1}{\sqrt{3}}\right), 90°$

c. $90°, \cos^{-1}\left(\frac{1}{\sqrt{3}}\right), 0°$

d. $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right), 90°, 0°$

02. A half section of thin circular pipe of mass m and radius r is released from rest in the position shown. Pipe rolls without sliding. The change in potential energy of pipe after it has rolled through 180°

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a. $\frac{4mgr}{\pi}$ b. $\frac{2mgr}{\pi}$

c. mgr d. $\sqrt{2}mgr$

03. An ideal gas is initially at $p_1, v_1$ is expanded to $p_2, v_2$ Isothermally and then compressed adiabatically to the same volume $v_1$ and pressure $p_3$. If W is the net workdone by the gas in complete process. Which of the following is true?

a. $W > 0; p_3 > p_1$ b. $W < 0; p_3 > p_1$

c. $W > 0; p_3 < p_1$ d. $W < 0; p_3 < p_1$

04. A stone attached to one end of a string, is revolved around a stick so that string winds upon the stick and gets shortened. Which of the following is conserved?

a. Linear momentum

b. Kinetic energy

c. Angular momentum

d. all of these

05. Two wheels, each with four spokes are mounted one behind the other and aligned such that an observer sees only four spokes. At t = 0 minute, one wheel begins to spin at 5 revolutions / minute and the other at 4 revolutions/minute in same sense. Upto t = 1 minute, how many times the observer would see only four spokes?

a. 1 b. 2

c. 4 d. 8

06. The ratio of the times taken by an uniform solid sphere and a disc of the same mass and diameter to roll down from rest the same distance on an inclined plane is

a. $\sqrt{\frac{15}{14}}$ b. $\sqrt{\frac{14}{15}}$

c. $\sqrt{\frac{11}{12}}$ d. $\sqrt{\frac{12}{11}}$

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07. Which of the following is correct

a. An astronaut in going from earth to moon will experience weightlessness once

b. When a thin uniform spherical shell gradually shrinks, the gravitational potential at its centre decreases

c. In the case of spherical shell, the plot of gravitational potential V versus r is continuous

d. All of these

08. The orbital velocity of a satellite close to the surface of the earth is V. If this satellite starts orbting at an altitude of half the Earth's radius the orbital velocity will be

a. $\sqrt{\frac{2}{3}}V$ b. $\sqrt{\frac{2}{5}}V$

c. $\sqrt{\frac{2}{7}}V$ d. $\sqrt{\frac{2}{9}}V$

09. A light rod of length 2m is suspended from the ceiling horizontally by means of two vertical wires of equal length to its end. One of the wires is made of steel and is of cross-section $0.1cm^2$.The other wire is of brass of cross-section $0.2cm^2$. A weight is suspended from a certain point of the rod such that equal stresses are produced in both the wires. Which of the following is correct?

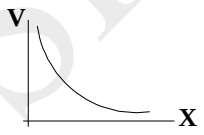
a. the ratio of tension in the steel and brass wires is 2:1

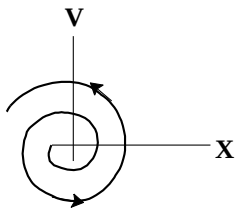
b. the load is suspended at a distance of $\frac{200}{3}$ cm from the steel wire

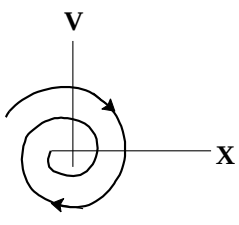
c. Both a and b are correct

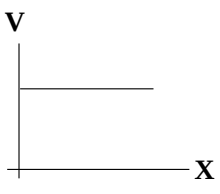
d. Both a and b are wrong

10. In damped oscillation ; graph between velocity and position will be

a. 


b. 

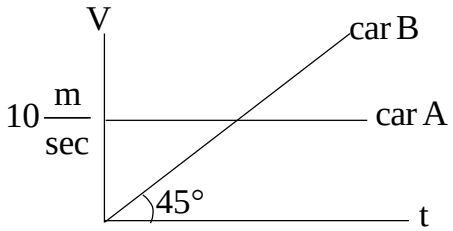

c. 


d. 




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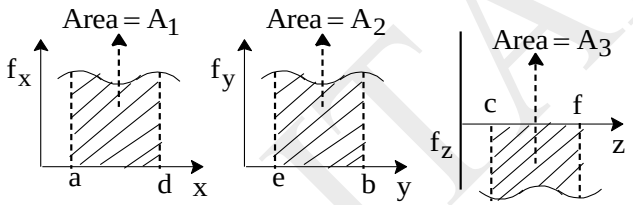
11. Initially car A is 10.5m a head of car B. Both start moving at time t = 0 in the same direction along a straight line. The velocity time graph of two cars is shown in figure. The time when the car B will catch the car A will be




a. t= 20 sec b. t = $2\sqrt{5}$ sec

c. t = 21 sec d. t = 18 sec

12. A force given by $\vec{F} = f_x\hat{i} + f_y\hat{j} + f_z\hat{k}$ acts on a particle which moves from (a, b,c) to (d, e, f). The work done by force F is (here $A_1, A_2$ and $A_1, A_3$ magnitude of area bounded)




a. $A_1 + A_2 + A_3$ b. $A_1 - A_2 - A_3$

c. $-A_1 + A_2 - A_3$ d. $A_1 - A_2 + A_3$

13. An earthen pitcher loses 1g of water per minute due to evaporation. If the water equivalent of the pitcher is 0.5kg and the pitcher contains 9.5kg of water, then calculate the time required for the water in pitcher to cool to 28°C from its original temperature of 30°C. Neglect the effect of radiation. Latent heat of vaporisation of water in this range of temperature is 580 cal/g and specific heat of water is 1cal /g/°C

a. 30.5 min b. 41.2 min

c. 38.6 min d. 34.5 min

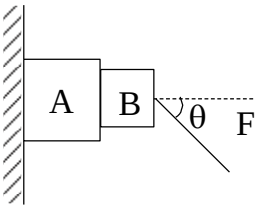
14. A screen is placed 90cm from an object. The image of an object on the screen is formed by a convex lens at two different location separated by 20cm. The focal length of the lens is

a. 18 cm b. 21.4 cm

c. 60 cm d. 85.6 cm

15. Consider the situation shown in the figure. The wall is smooth but the surfaces of blocks A and B in contact are rough. The friction on B due to A

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a. Is upward b. Is downward

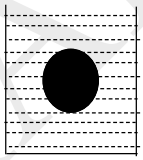
c. Is zero d. none of these

16. A massive stone pillar 20m high and of uniform cross-section rests on a rigid base and supports a vertical load of $5\times10^5$ N at its upper end. If the compressive stress in the piller is not exceed $1.6\times10^6 Nm^{-2}$. What is the minimum cross-sectinal area of the pillar? Density of the stone = $2.5\times10^3 Kg/m^3$ (Take g = 10N $Kg^{-1}$)

a. $0.15m^2$ b. $0.25m^2$

c. $0.35m^2$ d. $0.45m^2$

17. The system shown in figure is allowed to fall freely under gravity. The upthrust on the immersed body is



a. zero b. infinite

c. equal to weight of liquid displaced

d. greater than weight of liquid displaced

18. A box (thermally insulated) has two chambers separated by a membrane. One of volume V contains an ideal gas at temperature T. The other of volume V/2 is evacuated. If the membrane breaks down, the gas temperature will be

a. T/2 b. 3T/2

c. T d. 2T

19. 16g of oxygen and 14g of nitrogen are mixed in an enclosure of volume $5000cm^3$ at a temperature of 300K. The resulting pressure of the mixture is nearly

a. $5\times10^5$ Pa b. $4\times10^5$ Pa

c. $3\times10^5$ Pa d. $1.2\times10^5$ Pa

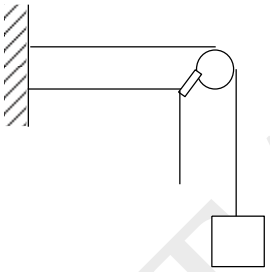
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20. The temperature of a furnace is 2324°C and the intensity is maximum in its radiation spectrum nearly at 12000Å. If the intensity in the spectrum of a star is maximum nearly at 4800Å, then the surface temperature of star

a. 8400°C b. 7200°C

c. 6219°C d. 5900°C

21. Fig shows a string stretched by a block going over a pulley. The string vibrates in its tenth harmonic in union with a particular tuning fork. When a beaker containing water is brought under the block so that the block is completely dipped into the beaker, the string vibrates in its eleventh harmonic. The density of the material of the block is



a. $1.0\times10^{3}\text{Kgm}^{-3}$ b. $3.0\times10^{3}\text{Kgm}^{-3}$

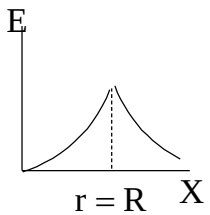
c. $5.0\times10^{3}\text{Kgm}^{-3}$ d. $5.8\times10^{3}\text{Kgm}^{-3}$

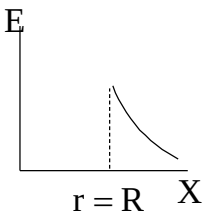
22. A piece of wood has dimensions a,b and c. Its relative density is d. It is floating in water such that the side a is vertical. Its is now pushed down gently and released. The time period is

a. $2\pi\sqrt{\frac{abc}{g}}$ b. $8\pi\sqrt{\frac{bc}{dg}}$

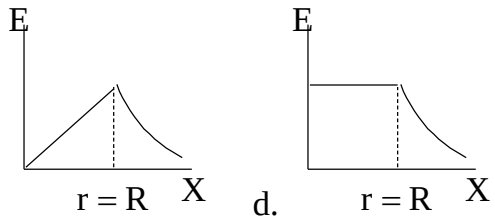
c. $2\pi\sqrt{\frac{g}{ab}}$ d. $2\pi\sqrt{\frac{ad}{g}}$

23. A hollow charged sphere of radius R has a constant surface charge density $\sigma$. Of the following graphs the one, which shows the variation of the electric field strength E with distance X from the centre of the sphere is

a. 


b. 


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c. d.

24. A point charge of $5\times10^{-9}\text{C}$ is transferred from infinity to a point 5cm from the surface of a ball of radius 5cm and charge density $5\times10^{-4}\text{C/m}^3$. The amount of work done in the process is

a. $0.12\times10^{-3}\text{J}$ b. $7.07\times10^{-6}\text{J}$

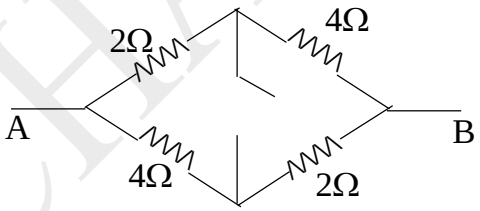
c. $7.07\times10^{-4}\text{J}$ d. 70 J

25. Two parallel metal plates carry charges $+\text{Q}$ and $-\text{Q}$. A charge placed between the plates experience a force F. The plates are then moved apart so that the separation between them is doubled. The force on the test charge will now be

a. F/2 b. F

c. F/4 d. 2F

26. What are the equivalent resistance $\text{R}_{\text{AB}}$ between the points A and B in the network shown in figure



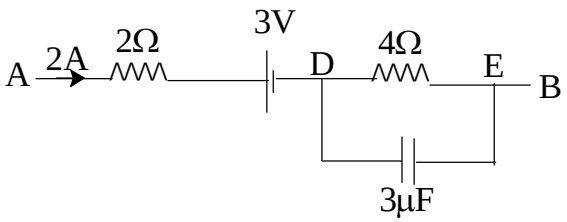

a. when switch S is open, $\text{R}_{\text{AB}}=\frac{8}{3}\Omega$

b. when switch S is open, $\text{R}_{\text{AB}}=8\Omega$

c. When switch S is closed $\text{R}_{\text{AB}}=9\Omega$

d. When switch S is closed $\text{R}_{\text{AB}}=\frac{8}{3}\Omega$

27. What is the energy stored in the capacitor is



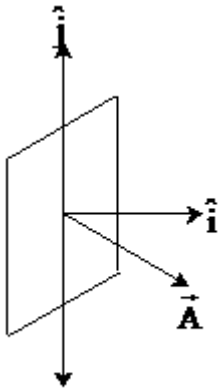



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a. $72\,\mu J$ b. $96\,\mu J$

c. 96 mJ d. 96 MJ

28. The torque acting on a rectangular loop having area A carrying current i and placed in a magnetic field $B\hat{i}$, with which the loop makes an angle $\theta$ is given by




a. $BiA\hat{i}$ b. $BiA\sin\theta\hat{k}$

c. $BiA\cos\theta\hat{j}$ d. $BiA\tan\theta\hat{k}$

29. A beam of protons with a velocity of $4\times10^5 ms^{-1}$ enters a uniform magnetic field of 0.3T. The velocity makes an angle of 60° with the magnetic field. Find the radius of the helical path taken by the proton beam and the pitch of the helix

a. 4.4 cm and 1.2 cm

b. 1.2 cm and 4.4 cm

c. 4.4 cm and 4.4 cm

d. 1.2 cm and 1.2 cm

30. Induction coil is based on the principle of

a. rheostat b. self induction

c. mutual induction d. none of these

31. The displacement current exist whenever the following quantity changes with time

a. Charge b. Magnetic field

c. electric flux d. magnetic flux

32. Imagine an atom made of a proton and a hypothetical particle of double the mass of electron but having same charge as the electron. Apply Bohr atom model and consider all possible transition of the hypothetical particle to the first excited level. The longest

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wavelength photon that will be emitted has wavelength $\lambda$ (in Rydberg constant terms) equal to

a. 9/5R b. 54/5R

c. 18/5R d. 4/R

33. A short bar magnet with the north-pole facing north forms a neutral point at point P in the horizontal plane. If the magnet is rotated by 90° in the horizontal plane, the net magnetic induction at P is (horizontal component of earths magnetic field = $B_H$)

a. zero b. $2B_H$

c. $\frac{\sqrt{5}}{2}B_H$ d. $\sqrt{5}B_H$

34. The average value of electric energy density in an electromagnetic wave is ($E_0$ is peak value of electric field in electromagnetic wave)

a. $\frac{1}{2}\in_0 E_0^2$ b. $\frac{E_0^2}{2\in_0}$

c. $\in_0 E_0^2$ d. $\frac{1}{4}\in_0 E_0^2$

35. K.E of photon electrons is E. When light of wavelength $\lambda$ falls on them. If K.E of photon electrons reduces to half i.e. E/2, then, wavelength of incident light will be

a. $\frac{hc\lambda}{E\lambda - hc}$ b. $\frac{hc(2hc - E\lambda)}{2\lambda}$

c. $2\lambda$ d. $\frac{\lambda}{2}$

36. A glass sphere of radius $5\times10^{-2}$ m has a small bubble $2\times10^{-2}$ m from its centre. Bubble is viewed along the diameter of the sphere, from the side on which it lies. If refractive index of glass is 1.5, then how far from the surface will the bubble appear?

a. 2.1 cm b. 2.5 cm

c. 1.5 cm d. 2.0 cm

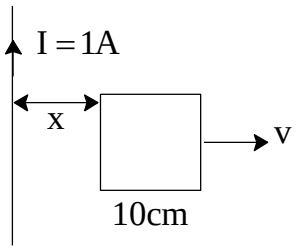
37. The focal length of a mirror in air is f. The focal length of the same mirror if in a liquid of refraction index $\sqrt{2}$ is

a. f b. $\sqrt{2}$ f

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c. $\frac{f}{\sqrt{2}}$ d. 2f

38. A square frame of side 10cm carrying current 1A and a long straight wire carrying current 1A are in the plane of paper. Starting from close to the wire. The frame moves towards the right with a constant speed of $10ms^{-1}$. The e.m.f. induced at the time the left arm of the frame is at x = 10cm from the wire is




a. 2μv b. 1μv

c. 0.75μv d. 0.5μv

39. Current of 10 ampere and 2 ampere are passed through two parallel thin wires A and B respectively in opposite directions. Wire A is infinitely long and the length of the wire B is 2m. The magnetic force acting on the conductor B which is situated at 10cm distance from A will be

a. $8\times10^{-5}N$ b. $5\times10^{-5}N$

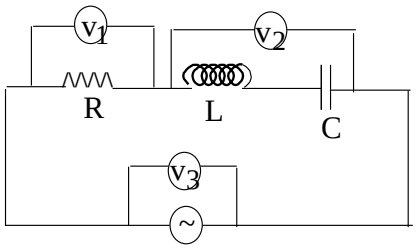
c. $8\pi\times10^{-7}N$ d. $4\pi\times10^{-7}N$

40. N divisions on the main scale of a vernier callipers coincides with (N+1) divisions of the vernier scale. If each division of main scale is ‘a’ units, then the least count of the instrument is

a. a b. $\frac{a}{N}$

c. $\left(\frac{N}{N+1}a\right)$ d. $\frac{a}{N+1}$

41. In the figure shown, three ac voltmeter have been connected. At resonance the reading of



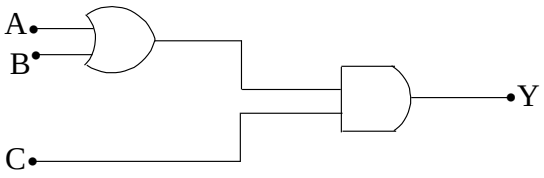



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a. $v_1$ is zero b. $v_2$ is zero

c. $v_3$ is zero d. $v_1 = v_2 = v_3 = 0$

42. In order to obtain an output Y=1 from the circuit of fig, the inputs must be




a. A = 0, B=1 and C = 0

b. A = 1, B=0 and C = 1

c. A = 1, B=0 and C = 0

d. A = 1, B=1 and C = 0

43. Half lives of two radioactive substances A and B are respectively 20 minutes and 40 minutes. Initially the sample of A and B have equal number of nuclei. After 80 minutes the ratio of the remaining number of A and B nuclei is

a. 1 : 16 b. 4 : 1

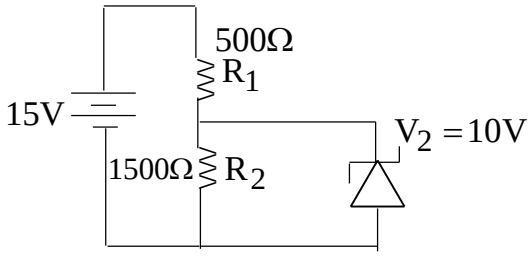
c. 1 : 4 d. 1 : 1

44. Two tangent galvanometer having coils of the same radius are connected in series. A current flowing in them produces deflections of 60° and 45° respectively. The ratio of the number of turns in the coil is

a. $4/3$ b. $(\sqrt{3}+1)/1$

c. $(\sqrt{3}-1)/1$ d. $\sqrt{3}/1$

45. In the given circuit, the current through the zener diode is




a. 2.5 mA b. 3.3 mA

c. 5.5 mA d. 6.7 mA

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**CHEMISTRY**

46. Which of the following is an anionic detergent?

a. Sodium lauryl sulphate

b. Cetyltrimethyl ammonium bromide

c. Glyceryl oletate

d. Sodium stearate

Ans. a

Sol. In sodium lauryl sulphate, anions associate to form micelles

47. The number of 2-centre-2-electron and 3-centre -2-electron bonds in $B_2H_6$, respectively, are

a. 4 and 2 b. 2 and 4 c. 2 and 2 d. 2 and 1

Ans. a

Sol. $4B-H_T$ have 2C – 2e bonds and two B – H – B bridges have 3C – 2e bonds

48. The main oxides formed on combustion of Li, Na and K in excess of air respectively are

a. $LiO_2, Na_2O_2$ *and* $K_2O$

b. $Li_2O_2, Na_2O_2$ *and* $KO_2$

c. $Li_2O, Na_2O_2$ *and* $KO_2$

d. $Li_2O, Na_2O$ *and* $KO_2$

Ans. c

Sol. Li mainly forms oxide

Na can form peroxide in excess oxygen

K, Rb, Cs form super oxides

49. $2Fe(s)+O_2(g)+4H^+(aq)\longrightarrow 2Fe^{+2}(aq)+2H_2O(\ell)$.If $E_{cell}=E^o_{cell}, \left[Fe^{+2}\right]=10^{-3}M, P_{O_2}=0.01atm, P^H=x$. '$x$' is

a. 2 b. 1

c. 4 d. 3

Ans. b

Sol. $E_{cell}-E^o_{cell}=0$



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$$0 = \frac{-0.0591}{4}\log\frac{\left[Fe^{+2}\right]^2}{\left[H^+\right]^4\left(P_{O_2}\right)}$$

$$0 = \log\frac{\left(10^{-3}\right)^2}{\left[H^+\right]^4(0.01)}$$

$$1 = \frac{10^{-6}}{\left[H^+\right]^4(0.01)}$$

$$\left[H^+\right]^4 = 10^{-4}\,M$$

$$\left[H^+\right] = 10^{-1}\,M$$

$$P^H = 1$$

50. The hydride that is not electron deficient is

a. $AlH_3$ b. $B_2H_6$

c. $SiH_4$ d. $GaH_3$

Ans. c

Sol. $SiH_4$ is electron precise

51. Name the structure of silicates in which three oxygen atoms of $\left[SiO_4\right]^{4-}$ are shared is

a. pyrosilicate

b. sheet silicate

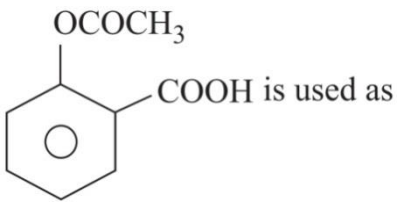
c. linear chain silicate

d. three – dimensional silicate

Ans. b

Sol. Conceptual, 2-D structure, $\left(Si_2O_5\right)_x^{2n-}$; Eg Mica, Talc etc

52.




is used as

a. Insecticide b. Antihistamine

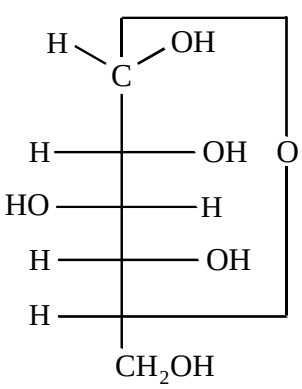
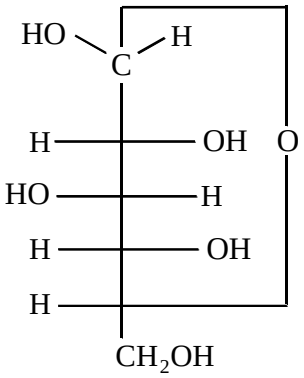
c. Analgesic d. Antacid



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Ans. c

Sol. Aspirin is analgesic

53.  and  are

a. Enantiomers

b. Diastereomers

c. Epimers

d. Functional isomers

Ans. b

Sol. They are anomers. Anomers are diastereomers

54. The number of stereoisomers obtained by addition of $Br_2/CCl_4$ to trans-2-butene is

a. 1 b. 2

c. 3 d. 4

Ans. a

Sol. $Br_2/CCl_4$ anti addition

Anti addition with trans 2 – butene gives meso product

55. The compound with the highest boiling point is

a. n-pentane

b. n-Hexane

c. 2-methyl butane

d. 2,2-Dimethyl propane

Ans. b

Sol. Higher the molecular weight, greater the BP

56. Wrong match about the uses of polymers is

a. Polystyrene ……. Wrapping material, insulator

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b. PVC ……. Rain coats, Hand bags

c. Bakelite ……. Substitute to wool

d. Glyptal ……. Paints and Lacquers

Ans. c

Sol. Bakelite is used in making electrical switches, handles to utensils, phonographic records, combs

57. $CH_3-CH=CH_2+HOCl \rightarrow P$ Identify the major product

a. $CH_3-\underset{\underset{OH}{|}}{CH}-\underset{\underset{Cl}{|}}{CH_2}$

b. $CH_3-\underset{\underset{Cl}{|}}{CH}-\underset{\underset{OH}{|}}{CH_2}$

c. $CH_3-CH_2-\underset{\underset{Cl}{|}}{CH}-OH$

d. $\underset{\underset{OH}{|}}{CH_2}-CH_2-\underset{\underset{Cl}{|}}{CH}$

Ans. a

Sol. $HOCl \longrightarrow Cl^+ + HO^-$

Negative part of addendum attacks unsaturated carbon with lesser number of hydrogens

58. Identify a reagent from the following list which can easily distinguish between 1 – butyne and 2-butyne

a. Bromine, $CCl_4$

b. $H_2$, Lindlar's catalyst

c. Dilute $H_2SO_4, HgSO_4$

d. Ammonical $Cu_2Cl_2$ solution

Ans. d

Sol. 1 – butyne reacts with $Cu_2Cl_2/NH_2OH$ but not 2 – butyne.

1 – butyne has acidic H - atoms

59. When phenyl magnesium bromide reacts with tertiary butyl alcohol, the product is

a. Benzene b. Phenol



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c. 3° Butyl benzene d. 3° Butyl phenyl ether

Ans. a

Sol.

$$C_6H_5MgBr + (CH_3)_3COH \longrightarrow C_6H_6 + Mg\begin{matrix} Br \\ OC(CH_3)_3 \end{matrix}$$

60. Butan-2-one to propanoic acid conversion occurs in presence of

a. Tollens reagent b. Fehlings solution

c. $NaOH / I_2 / H^+$ d. 2, 4 – DNP

Ans. c

Sol. Iodoform test, followed by acid hydrolysis $(CH_3CH_2COOH, CHI_3)$

61. When acetaldehyde is heated with Fehling's solution, it gives a precipitate of

a. Cu b. CuO

c. $Cu_2O$ d. $Cu + Cu_2O + CuO$

Ans. c

Sol. In Fehlings test, CuO reduced to $Cu_2O$ (Red ppt)

62. Chlorination of toluene in the presence of light and heat followed by the treatment with aqueous NaOH gives:

a. o -Cresol b. p-Cresol

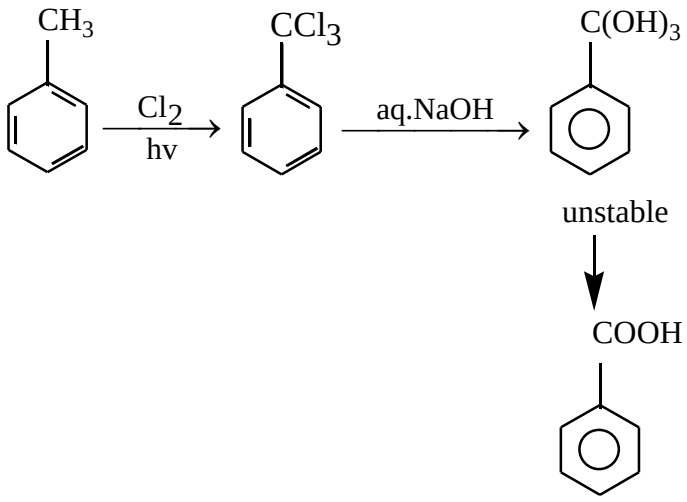
c. 2,4-dihydroxy toluene

d. Benzoic acid

Ans. d

Sol.

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63. Chlorobenzene can be prepared by reacting aniline with

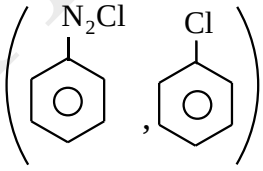
a. Hydrochloric acid

b. Cuprous chloride

c. Chlorine in the presence of anhydrous aluminium chloride

d. Nitrous acid followed by heating with cuprous chloride

Ans. d

Sol. Diazotisation followed by Sandmayers reaction 


64. I. 1,2-Dihydroxy benzene

II. 1,3-Dihydroxy benzene

III. 1,4-Dihydroxy benzene

IV. Hydroxy benzene

The increasing order of boiling points of the above mentioned list is

a. I < II < III < IV b. I < II < IV < III

c. IV < I < II < III d. IV < II < I < III

Ans. c

Sol. P-isomer has inter molecular hydrogen bond'

O-isomer has intramolecular hydrogen bond

65. When propionic acid is treated with aqueous sodium bicarbonate, $CO_2$ is liberated.

Carbondioxide comes from

a. Methyl group

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b. carboxylic acid group

c. Methylene group

d. Bicarbonate group

Ans. d

Sol. $CH_3CH_2COOH + NaHCO_3 \longrightarrow CH_3COONa + H_2O + CO_2$

66. Among the following, the strongest base is

a. $C_6H_5NH_2$

b. $p-NO_2C_6H_4NH_2$

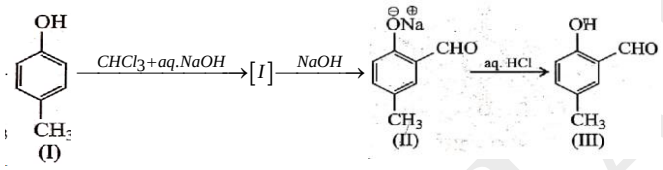
c. $m-NO_2.C_6H_4NH_2$

d. $C_6H_5CH_2NH_2$

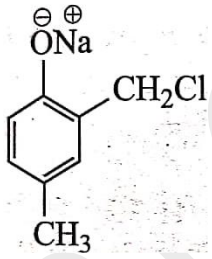
Ans. d

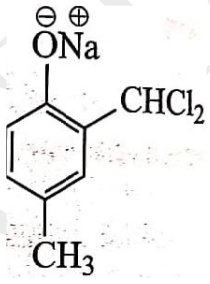
Sol. In Benzyl amine the lone pair is not involved in resonance

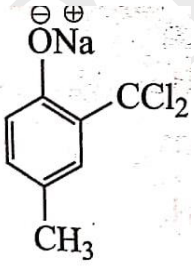
67.

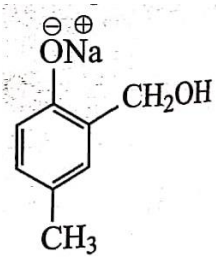



The structure of the intermediate (I) is:

a. 


b. 


c. 


d. 


Ans. b

Sol. Benzal chloride intermediate is formed

68. Zeotropic ( ideal) mixture of the following is

a. $C_6H_{14} + C_7H_{16}$

b. $CHCl_3 + CH_3COCH_3$

c. $C_2H_5OH + H_2O$

d. $H_2O + HNO_3$

Ans. a

Sol. An ideal solution is a zeotropic mixture

69. Freezing point of a biological fluid is $-0.62°C$ in aqueous solution $K_f(H_2O) = 1.86°mol^{-1}/kg$. The osmotic pressure of the solution at 300K is (assuming molarity = molality)

a. 8.2 atm b. 4.1 atm

c. 16.2 atm d. 2.8 atm

Ans. a

Sol. $\Delta T_f = K_f.m \Rightarrow m = \frac{0.62}{1.86} = \frac{1}{3}$

$\pi = MRT = \frac{1}{3} \times 0.082 \times 300 = 8.2\,atm$

70. A constant current of 30A is passed through an aqueous solution of NaCl for a time of 1.00 h, then mass of NaOH formed due to electrolysis is

a. 44.8g b. 31.1g

c. 22.4g d. 65.2g

Ans. a

Sol. Mass of NaOH = $\frac{w}{E} \times Eqwt = \frac{30 \times 3600}{96500} \times 40 = 44.8g$

71. For the reaction $A \rightarrow B$, rate constant (K) $= 10^{-3} Ms^{-1}$. If the initial concentration of A is 0.5M, the time needed for completion of reaction is

a. 1000s b. 500s

c. 250s d. 1500s

Ans. b

Sol. $t_{100\%} = \frac{(Initial\,Conc)}{K} = \frac{0.5}{10^{-3}} = 500s$ (zero order reaction)



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72. In acidic medium $KMnO_4$ oxidises oxalic acid to $CO_2$. The catalyst in the reactions is

a. $MnO_4^{2-}$ b. $Mn^{+2}$

c. $CO_2$ d. $CrO_4^{2-}$

Ans. b

Sol. Autocatalysis $(Mn^{+2})$

73. Which of the following shows reducing properties?

a. $H_3PO_2$ b. $H_3PO_3$

c. $H_3PO_4$ d. both a and b

Ans. d

Sol. Both $H_3PO_2$ and $H_3PO_3$ contain P-H bonds

74. Charring of cane sugar by concentrated $H_2SO_4$ is mainly due to

a. Oxidation

b. complex formation

c. Dehydration

d. strong acidic nature

Ans. c

Sol. $C_{12}H_{22}O_{11} \xrightarrow{\text{conc.}H_2SO_4} 12C + 11H_2O$

75. Chlorine gas is passed through NaOH in hot concentrated condition. The oxidation number of chlorine in the products

a. -1 and +1 b. -1 and +5

c. 0 and +5 d. 0 and +1

Ans. b

Sol. $3Cl_2 + 6NaOH \rightarrow \underset{(-1)}{5NaCl} + \underset{(+5)}{NaClO_3} + 3H_2O$

76. $XeF_6 + CsF \rightarrow \text{Product}$. Product is

a. $[CsF_2][XeF_5]$ b. $Cs[XeF_7]$

c. $[XeF_4][CsF_3]$ d. No reaction occurs

Ans. b

Sol. CsF is flouride ion donor



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77. A good reducing agent among the following is

a. $Mn^{+3}(aq)$ b. $Cr^{+2}(aq)$

c. $Zn^{+2}(aq)$ d. $Fe^{+3}(aq)$

Ans. b

Sol. $Cr^{+3}(aq)$ has stable $t_2g^3eg^0$ configuration, $E^o_{Cr^{+3}/Cr^{+2}}$ is negative

78. The number of stereo isomers possible for $[Co(en)_2Cl_2]^+$ are

a. 2 b. 3

c. 4 d. 8

Ans. b

Sol. one cis and one trans isomers are possible and cis isomer is optically active (d, l form)

79. Incorrect match is

a. Calamine - $ZnCO_3$

b. Siderite - $Fe_3O_4$

c. Sphalerite - $ZnS$

d. Malachite - $CuCO_3.Cu(OH)_2$

Ans. b

Sol. Siderite is $FeCO_3$

80. Element with high second ionisation enthalpy of the following

a. B b. C

c. N d. O

Ans. d

Sol. Stable Half filled configuration of $O^+(g)$

81. The number of molecules present in 112ml of He gas at 273°C and 2 atm are

a. $\frac{N_A}{100}$ b. $\frac{N_A}{200}$

c. $\frac{N_A}{500}$ d. $\frac{N_A}{250}$

($N_A$ = avogadro number)

Ans. b



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Sol. number of molecules = $\frac{112}{22400}\times N_A = \frac{N_A}{200}$ (STP), $\because$ temperature and pressure are doubled, volume remains constant

82. The standard enthalpy of formation is zero for

a. $Br_2(g)$ b. $H_2O(l)$

c. $S_8$ (monoclinic) d. $O_2(g)$

Ans. d

Sol. Standard state of oxygen is gas

83. If $\Delta G^0 = 0$ (zero) the equilibrium constant value of a reaction will be

a. K>1 b. K<1

c. K=1 d. K= 0

Ans. c

Sol. $\Delta G° = -2.303RT\log K$

As $\Delta G° = 0, K = 1$

84. The solubility of AgCl is maximum in

a. $H_2O$ b. $0.1M\,KCl(aq)$

c. $NH_3$ d. $0.05M AgNO_3(aq)$

Ans. c

Sol. $AgCl(s) + 2NH_3 \rightarrow [Ag(NH_3)_2]Cl$

85. Increasing skin cancer is mainly due to

a. Acid rain b. ozone depletion

c. CO pollution d. greenhouse gases

Ans. b

Sol. u.v radiation causes skin cancer

86. In chromium (Z =24), the number of electrons with $l$ =1 and $l$ = 2 are respectively

a. 12 and 4 b. 7 and 5

c. 12 and 5 d. 8 and 4

Ans. c

Sol. $l$ = 1(p – orbital) 12 electrons ($2p^6$and $3p^6$)

$l$ = 2(d – orbital) 5 electrons ($3d^5$)



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87. $xMnO_4^- + yH_2O_2 + zH^+ \rightarrow$

$Mn^{+2} + O_2 + H_2O$

Coefficients x, y, z are respectively in the balanced equation is

| | x | y | z |
|---|---|---|---|
| a. | 2 | 5 | 6 |
| b. | 2 | 6 | 5 |
| c. | 5 | 2 | 6 |
| d. | 3 | 2 | 6 |

Ans. a

Sol. $2MnO_4^- + 5H_2O_2 + 6H^+ \rightarrow 2Mn^{+2} + 5O_2 + 8H_2O$

88. B-F bond length in $BF_3$ is 130pm and in $BF_4^-$ is 143 due to

a. less electronegativity of fluorine in $BF_4^-$

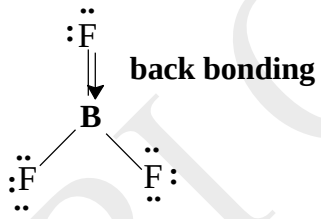
b. back bonding in $BF_3$

c. more electronegativity of boron in $BF_3$

d. tetrahedral shape of $BF_4^-$

Ans. b

Sol.




89. Amount of $NO(g)$ formed by oxidation of 1 mole of $NH_3(g)$ using 1mole of $O_2(g)$ is

a. 4.8g b. 56.0g

c. 30.0g d. 24.0g

Ans. d

Sol. $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$

One mole of $O_2$ reacts with 0.8 mole of $NH_3$ and forms 0.8 mole of NO

$(0.8 \times 30 = 24.0g)$

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90. In NaCl, chloride ions are in CCP arrangement, $Na^+$ ions are in octahedral voids. The ratio of $r_{Na^+} / r_{Cl^-}$ is

a. 0.225 b. 0.531

c. 0.845 d. 0.155

Ans. b

Sol. For 6 : 6 arrangement, radius ratio must lie between $0.414 \text{ to } 0.732$

## BOTANY

91. To which order solanaceae, convolvulaceae belongs

a. Rosales b. Polymoniales

c. Poales d. Sapindales

92. Which kingdom shows most extensive metabolic diversity?

a. Plantae b. Animalia

c. Fungi d. Monera

93. Identify the correct statements

a. Diatoms are chief producers in fresh water

b. *Gonyaulax* is a saprophytic protist with two unequal flagella right angles to each other

c. Mycoplasmas are obligate anaerobes

d. In slime moulds, plasmodium is formed under suitable conditions

94. *Ustilago, Claviceps, Alternaria, Colletotrichum, Puccinia, Albugo, Agaricus*

How many are Pathogenic fungi?

a. 6 b. 5

c. 4 d. 7

95. Anisogamous green alga is

a. *Fucus* b. *Spirogyra*

c. *Udorina* d. *Ulothrix*

96. In *Sphagnum* the position of antheridial branch and archegonial branch respecively

a. Upper, lower on same plant

b. Lower, upper on same plant

c. Separate plants

d. Both borne at upper side

97. Gametophyte is reduced and totally dependent in



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a. Algae, Angiosperms

b. Bryophytes, Pteriodophytes

c. Gymnosperms, Angiosperms

d. Gymnosperms, Bryophyte

98. Both fibrous roots and stilt roots found in

a. Banyan b. Maize

c. Sugar cane d. b and c

99. A lateral branch with short internodes and each node bearing a rosette of leaves and a tufts of balancing roots are found in

a. Pineapple, *Pistia*

b. *Eichhornia*, Banana

c. *Pistia*, Water hyacinth

d. Mint, *Chrysanthemum*

100. Replum is found in the ovary of

a. Mustard b. *Tridax*

c. Pea d. *Citrus*

101. Placentation, aestivation, bract side of flower can be observed in

a. Floral formula

b. Floral diagram

c. Both a & b

d. Achlamydeous flowers only

102. Shoot apical meristem forms

a. Axillary bud b. Leaf primordia

c. Intercalary meristem

d. More than one option is correct

103. The function of specialized parenchyma, closely associated with sieve tube elements helps in

a. Giving mechanical support to the plant

b. Translocation of water

c. Maintaining the pressure gradient in sieve cell

d. Maintaining the pressure gradient in sieve tube elements

104. Microbodies involved in glyoxalate cycle in germinating castor seed are



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a. Sphaerosomes b. Peroxisomes

c. Lomosomes d. Glyoxysomes

105. S-I - Cell sap concentration in the vacuole is higher than in cytoplasm

S-II – Cell sap contain water soluble photosynthetic pigment called anthocyanin

a. Both statements are correct

b. S-I is correct S-II is incorrect

c. S-I is incorrect S – II is correct

d. Both statements are wrong

106. Biomolecule whose molecular weight. does not exceeds 800 daltons is

a. Proteins b. Nucleic Acids

c. Lipids d. Polysaccharides

107. Identify the correct combination

a. Insulin – Polymer of aminoacids

b. Chitin – Heteropolysaccharide

c. Ribozyme – Non- proteinaceous enzyme

d. More than one option is correct

108. Separation of homologous chromosomes from bivalents towards opposite poles occur at

a. Anaphase - II b. Anaphase

c. Anaphase - I d. a and b

109. Kinases + CdK controls

a. Cell cycle b. DNA

c. Centriole duplication d. Cytokinesis

110. For a solution at atmospheric pressure, water potential is equal to

a. $\psi_s$ b. $\psi_p$

c. $\psi_m$ d. 0

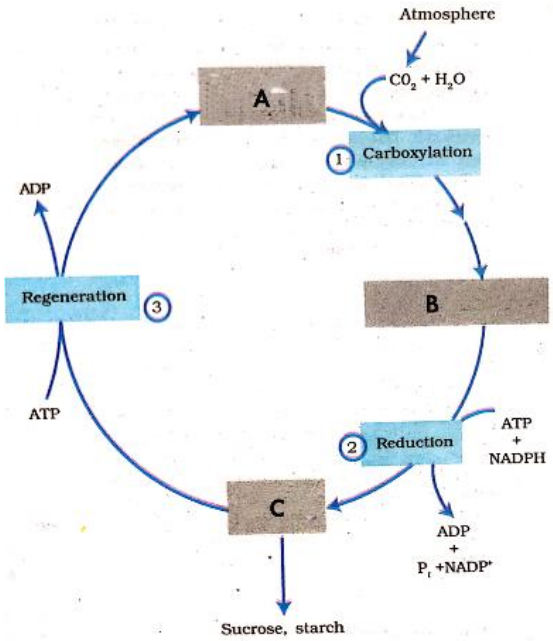
111. X-Maintains the ribosome structure, Y-Essential for formation of chlorophyll, Z - Required for pollen germination respectively

a. Mg, Fe, B b. Mn, Fe, Zn

c. Mo, Cl, Zn d. K, Fe, Ca

112. Identify the A,B,C respectively from the given below diagram of Calvin cycle.

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| | A | B | C |
|---|---|---|---|
| a) | Triose phosphate | Ribulose 1,5-bisphosphate | 3-Phosphoglycerate |
| b) | Ribulose 1,5-bisphosphate | 3-Phosphoglycerate | Triose Phosphate |
| c) | 3-Phosphoglycerate | Ribulose 1,5-bisphosphate | Triose Phosphate |
| d) | Triose Phosphate | 3-Phosphoglycerate | Ribulose 1,5-bisphosphate |

113. Semi - autonomous organelles

a. Contain electron carriers

b. Contain porins

c. Shows chemiosmosis

d. More than one option is correct

114. Site of substrate level photosphorylation in plant cell

a. Chloroplast, Mitochondrial cristae

b. Cytoplasm, Perimitochondrial space

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c. Stroma of chloroplast and Cytoplasm

d. Cytosol, Mitochondrial matrix

115. Identify the incorrect pair

a. Cytokinins – Richmond – Lang effect

b. $GA_3$ – Triple response growith

c. ABA –Stress hormone

d. Auxin- Induces apical dominance

116. Internal, resting buds formed in

a. Sponges – Gemma

b. Sponges – Gemmules

c. *Penicillium* – Conidia

d. *Agave* - Bulbils

117. Ploidy of perisperm in pepper and beet root plant is

a. 2n b. 3n

c. n d. 4n

118. Generally a megaspore mother cell in the ovule differentiates in

a. Chalazal region b. Centre

c. Micropylar region

d. Hilum

119. Who used the frequency of recombination between gene pairs on the same chromosomes as a measure of distance between genes and mapped their position on the chromosomes?

a. T.H.Morgan b. Sutton

c. Alfred Sturtevant

d. Boveri

120. Both phenotypic and genotypic ratios are 1:2:1 in $F_2$ generation are found in

a. Mendels monohybrid cross

b. Correns incomplete dominance in snap dragon

c. Mendels dihybrid cross

d. Both a and b

121. Avery et.al, conducted biochemical character of transforming principle in

a. 1856-1863 b. 1833-1844

c. 1933 -1944 d. 1900

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122. Enzyme that catalyses teminism is

a. DNA dependent RNA polymerase

b. RNA dependent RNA polymerase

c. DNA dependent DNA polymerase

d. RNA dependent DNA polymerase

123. Identify the X in the given below diagram?




a. Tyrosine b. Serine

c. Phenylalanine d. Tryptophan

124. Identify the correct pair

a. Rapeseed mustard – Pusa Sawani

b. Cauliflower – Pusa Snowball K-1

c. Flat bean – Pusa $A_4$

d. Chilli – Karan rai

125. Anaerobic sludge digester releases a set of gases

a. $CO_2, CH_4, NO_3^-$ b. $CO, CH_4, H_2S$

c. $CO_2, CH_4, N_2O$ d. $CO_2, CH_4, H_2S$

126. Heterotrophic eukaryotic biofertilizer is

a. *Methylophilous* b. *Spirullina* c. *Glomus* d. *Anabaena*

127. $20^{th}$ century off –shoot of modern biology is

a. Biochemistry b. Bioinformatics c. Biotechnology d. Bioremediation

128. In pBR322 vector which REN has cloning site in rop gene.

a. *Pvu* - I b. *Pvu*-II c. *Pst* - I d. *Sal* - I

129. Separation and purification of recombinant protein after biosynthetic process is known as

a. upstream processing

b. polymerase chain reaction

c. Gel electrophoresis

d. Down stream processing

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130. Source of β – carotene gene in golden rice variety is

a. Daffodil b. Dog flower c. *Erwinia* d. a and c

131. Out of the total cost of various ecosystem services, select the wrong combination

a. Soil formation – Accounts for 50%

b. Both recreation and nutrient cycling – Together accounts for less than 10%

c. Climate regulation – About 6%

d. Habitat for wild life – About 6%

132. Primary succession occur in

a.Newly cooled lava b. Newly created reservoir

c. Bare rock d. All of these

133. Corn borer is controlled by

a. *cry* IAc b. *cry* IIAb c. *cry* IAb d. *cry* IAc and *cry* IIAb

134. The female sex organ of *Chara* is

a.Archegonium b. Oogonium c. Antheridium d. Androecium

135. Match the Column A with Column B

| | Column –A | | Column –B |
|---|---|---|---|
| I) | Cork | A) | Phellogen |
| II) | Cork cambium | B) | Periderm |
| III) | Secondary cortex | C) | Phelloderm |
| IV) | Bark | D) | Periderm + Bast |
| | | E) | Phellem |

| | I | II | III | IV |
|---|---|---|---|---|
| a) | E | D | C | B |
| b) | E | A | C | D |
| c) | A | B | C | D |
| d) | E | B | D | A |

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# ZOOLOGY

136. Choose the statement that is related to the phylum having spongocoel

a. They have choanocytes or collar cells in the body

b. They have combplates

c. They are bilaterally symmetrical

d. All are marine in habitat

137. Hard and non-pliable ground substance is found in

a. areolar tissue b. cartilage

c. bone d. adipose tissue

138.Tight junctions

a. stop leakage of substances across a cell

b. cement neighbouring cells together

c.are communication junctions

d. allow diffusion of substances across the cells

139 .Which of the following is an oviparous animal?

a. Scoliodon b. Panthera

c. Neophron d. Canis

140. Bidder's canal is found in

a. Testis of male frog

b. Testis of humans

c. Kidney of male frog

d. Kidney of female frog

141. Choose the correct match for cockroach

a. Malpighian tubules – 1000-1500 in number

b. Spiracles – 20 in number

c. Anal cerci – 4 in number

d. Mushroom gland – 1 pair in number

142. In Pheretima, spermathecal apertures are

a. 4 pairs from $5^{th}$- $9^{th}$ segment

b. 8 pairs from $6^{th}$-$9^{th}$ segment

c. 4 pairs from $6^{th}$-$9^{th}$ segments

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d. 2 pairs from $5^{th}$ -$9^{th}$ segments

143. In human dentition, the tooth which are 3 rooted are

a. all lower molars

b. all upper molars

c. all molars

d. all upper premolars

144. In which parts of the human alimentary canal absorption of drugs or medicines occur?

a. Mouth only

b. Stomach only

c. Small intestine only

d. Mouth, stomach and large intestine

145. Functional residual capacity is

a. ERV +TV b. ERV + RV

c. IRV + RV d. TV + IRV

146.The $PO_2$ in tissues is

a. equal to $PO_2$ in pulmonary artery

b. equal to $PCO_2$ in alveoli of Lungs

c. equal to $PCO_2$ in pulmonary vein

d. all the above

147.The following statement is related to the nucleated formed elements of blood which are largest in number

a. They are phagocytic in nature

b.They have haemoglobin

c.They cause allergic reactions

d.They are formed from megokaryoblasts

148. In human heart the AV valves are

a. 3 tricuspid valves and 2 bicuspid valves

b. 3 tricuspid and 2 bicuspid valve

c. 3 tricuspid valves and a bicuspid valve



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d. a tricuspid and a bicuspid valve

149. A condition in which the heart stops beating is

a. Angina pectoris b. Cardiac arrest

c. Heart failure d. Heart attack

150. Land snails are

a. Ammonotelic b. Uricotelic

c. Ureotelic d. Aminotelic

151. In the DCT of the nephron

a. conditional reabsorption of $Na^+$ and water takes place

b. the filtrate becomes hypotonic

c. urea undergoes reabsorption

d. glucose is reabsorbed

152. Choose the vasoconstrictor from the following

a. ANF b. ADH

c. ACTH d. PRL

153. Excretion of lactic acid from the body takes place by

a. Kidney b. Lungs

c. Sebaceous glands d. Sweat glands

154. In the myofibril of a skeletal muscle

a. I band has only thin filaments

b. A band has both thin and thick filaments

c. Thin filaments slide over thick filaments

d. All the above statements are true

155. Total number of bones included in skull is

a. 29 b. 22

c. 26 d. 23

156. Choose the incorrect statement

a. Rib cage has 37 bones

b. Sternum is ventral flat bone

c. Ribs are double headed ventrally

d. Scapula is large triangular bone

157. Select the statement(s) related to Schwaan cells.



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a. They are glial cells of PNS

b. They are found wrapped around axon to from myelin sheath

c. They are absent in non myelinated neurons

d. Both a and b are true

158. Corpus callosum connects

a. the two cerebral hemispheres

b.the two cerebellar hemispheres

c. mid brain and hind brian

d. diencephalon and medulla oblongata

159. Hyper calcemic hormone is

a. Parathormone b. Calcitonin

c. Insulin d. Glucagon

160. Hormone that has anabolic effect on carbohydrates is

a. Oestrogen b.Oxytocin

c. Ctrayicoc.exe laxin

161. The off site conservation of animals is done at

a. Sanctuaries

b. Zoological parks

c. National parks

d. Biosphere reserves

162. Find out the exotic species to Indian ecosystem

a. Lantana

b. Clarias gariepinus

c. Nile perch

d. Both a and b

163. According to Alexander Humboltz, the value of Z is 1.15 for

a. plants in Britain

b. fruit eating birds and mammals in tropical forests

c. molluscans in New York

d. birds in California

164. Cocaine is

a. stimulant

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b. hallucinogen in lower dose

c. depressant

d. bhang

165. A non – ionising carcinogen is

a. X - rays b. UV rays

c. Gamma rays d. IR rays

166. Drugs that can be used to treat allergy are

a. Anti histamines

b. Analgesics

c. Anti steroid drugs

d.Amphetamines

167.Colostrum from the mother provides….to the child

a. Natural active immunity

b. Natural passive immunity

c. Artificial active immunity

d. Artificial passive immunity

168.A graft rejection is due to

a.cellmediated immunity

b.auto immune disorder

c. humoral immunity

d. both b and c

169. Choose the incorrect statement

a. Salmonella causes typhoid

b. Gametocytes of Plasmodium are formed in RBC of man

c. Microsporum causes ringworm infection

d. Ascariasis is an air – borne infection

170. The ancestors of mammals are

a. Thecodonts b.Therapsids

c. Dinosaurs d. Pterosaurs

171. In stabilizing selection,

a. the average character value is removed

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b. the average character value is selected

c. individuals acquire peripheral character value

d. average character value shifts in one direction

172. The convergent form of Lemur is

a. Numbat b. Bobcat

c. Wolf d. Spotted cuscus

173. Choose the correct statement

a. GFC is major conduit for energy flow in a terrestrial ecosystem

b. DFC is major conduit for energy flow in an aquatic ecosystem

c. Standing crop can be measured as biomass or number in a unit area

d. DFC begins with producers

174. The Air Act was amended in

a. 1981 b. 1987

c. 1986 d. 2010

175. Catalytic converters do not have metals like

a. Platinum b. Palladium

c. Rhodium d. Lead

176. The following involves IVF except

a. ICSI b. ZIFT

c. GIFT d. IUT

177. Lippe's loop is a/an

a. Barrier for males

b. Implant

c. IUD

d. Female condom

178. Find out the number of spermatozoa obtained from 450 primary spermatocytes

a. 450 b. 900 c. 1200 d. 1800

179. Foetal ejection reflex is

a. generated by fully formed foetus

b. generated by placenta

c. a neuro endocrine mechanism

d. all the above are true

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180. In humans, karyotype AAXXY is not

a. exhibiting gynaecomastism
b. a sterile person
c. Klinefelter syndrome
d. Turner syndrome

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*Sec :* *Date :*

# SRI CHAITANYA EDUCATIONAL INSTITUTIONS,INDIA.

**A.P,TELANGANA,KARNATAKA,TAMILNADU,MAHARASHTRA,DELHI,RANCHI**

**SEC :Out going srs and longterm** **DATE: 24.07.2020**

**SUB:** **NEET GRAND TEST-14** **Max. Marks : 720**

## PHYSICS

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| *1)* b | *2)* b | *3)* b | *4)* c | *5)* c | *6)* b | *7)* d | *8)* a | *9)* d | *10)* c |
| *11)* c | *12)* b | *13)* d | *14)* b | *15)* b | *16)* d | *17)* a | *18)* c | *19)* a | *20)* c |
| *21)* d | *22)* d | *23)* b | *24)* a | *25)* b | *26)* d | *27)* b | *28)* c | *29)* b | *30)* c |
| *31)* c | *32)* c | *33)* d | *34)* d | *35)* b | *36)* b | *37)* a | *38)* b | *39)* a | *40)* d |
| *41)* b | *42)* b | *43)* c | *44)* d | *45)* b | | | | | |

## CHEMISTRY

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| *46)* a | *47)* a | *48)* c | *49)* b | *50)* c | *51)* b | *52)* c | *53)* b | *54)* a | *55)* b |
| *56)* c | *57)* a | *58)* d | *59)* a | *60)* c | *61)* c | *62)* d | *63)* d | *64)* c | *65)* d |
| *66)* d | *67)* b | *68)* a | *69)* a | *70)* a | *71)* b | *72)* b | *73)* d | *74)* c | *75)* b |
| *76)* b | *77)* b | *78)* b | *79)* b | *80)* d | *81)* b | *82)* d | *83)* c | *84)* c | *85)* b |
| *86)* c | *87)* a | *88)* b | *89)* d | *90)* b | | | | | |

## BOTANY

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| *91)* **b** | *92)* **d** | *93)* **d** | *94)* **a** | *95)* **c** | *96)* **a** | *97)* **c** | *98)* **d** | *99)* **c** | *100)* **a** |
| *101)* **b** | *102)* **d** | *103)* **d** | *104)* **d** | *105)* **b** | *106)* **c** | *107)* **d** | *108)* **c** | *109)* **a** | *110)* **a** |
| *111)* **a** | *112)* **b** | *113)* **d** | *114)* **d** | *115)* **b** | *116)* **b** | *117)* **a** | *118)* **c** | *119)* **c** | *120)* **b** |
| *121)* **c** | *122)* **d** | *123)* **b** | *124)* **b** | *125)* **d** | *126)* **c** | *127)* **c** | *128)* **b** | *129)* **d** | *130)* **d** |
| *131)* **b** | *132)* **d** | *133)* **c** | *134)* **b** | *135)* **b** | | | | | |

## ZOOLOGY

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| *136)* **a** | *137)* **c** | *138)* **a** | *139)* **c** | *140)* **c** | *141)* **b** | *142)* **a** | *143)* **b** | *144)* **d** | *145)* **b** |
| *146)* **d** | *147)* **a** | *148)* **d** | *149)* **b** | *150)* **b** | *151)* **a** | *152)* **b** | *153)* **d** | *154)* **d** | *155)* **a** |
| *156)* **c** | *157)* **d** | *158)* **a** | *159)* **a** | *160)* **b** | *161)* **b** | *162)* **d** | *163)* **b** | *164)* **a** | *165)* **b** |
| *166)* **a** | *167)* **b** | *168)* **a** | *169)* **d** | *170)* **b** | *171)* **b** | *172)* **d** | *173)* **c** | *174)* **b** | *175)* **d** |
| *176)* **c** | *177)* **c** | *178)* **d** | *179)* **d** | *180)* **d** | | | | | |